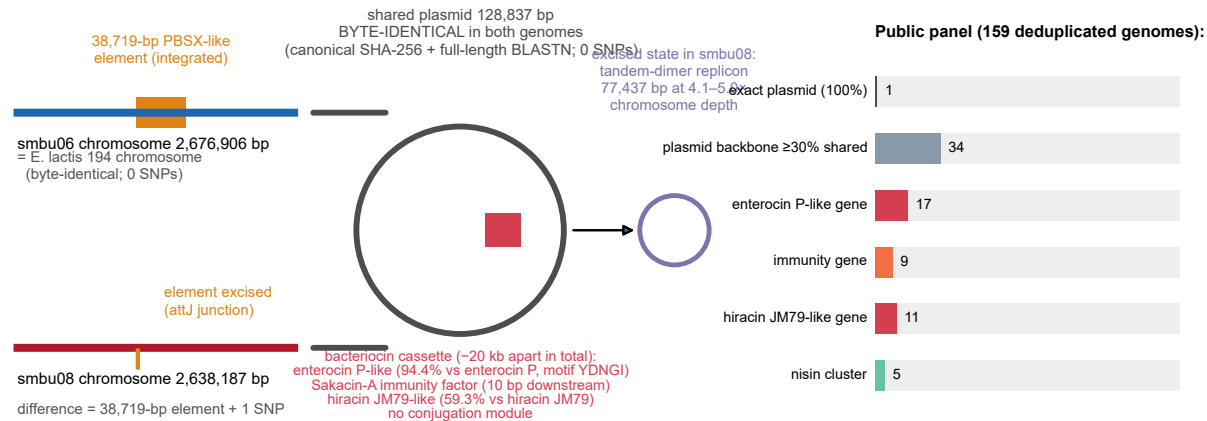


Graphical abstract



5/159 nisin clusters = the deliberately added controls only

OPEN HYPOTHESIS (not tested here): inter-plasmid/backbone mobilisation of the cassette, and any direction or timing of transfer — require population data and cannot be inferred from these static genomes.